

# Image Enhancement in the Spatial Domain

## Intensity Transformation

# Enhancement

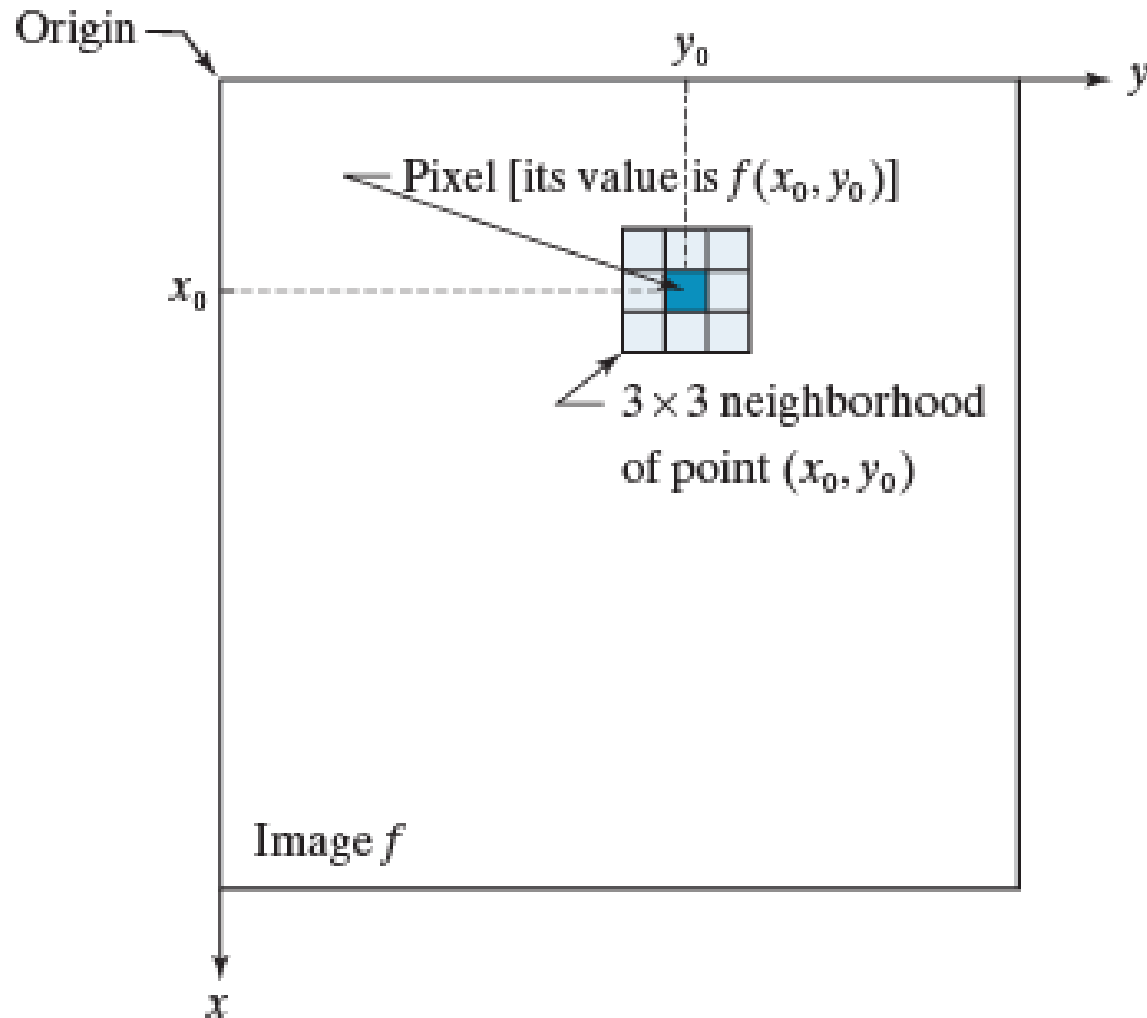
- Objectives
  - It makes an image more suitable for specific, problem-oriented applications than the original.
  - It is often for human perception; the viewer is the ultimate judge of how well a particular works.
- Two broad categories of enhancement methods
  - Spatial domain (this topic)
    - Works with pixels
  - Frequency domain (will be covered in the other topic)
    - Works with frequencies

# Spatial Domain

$$g(x,y) = T[ f(x,y) ]$$

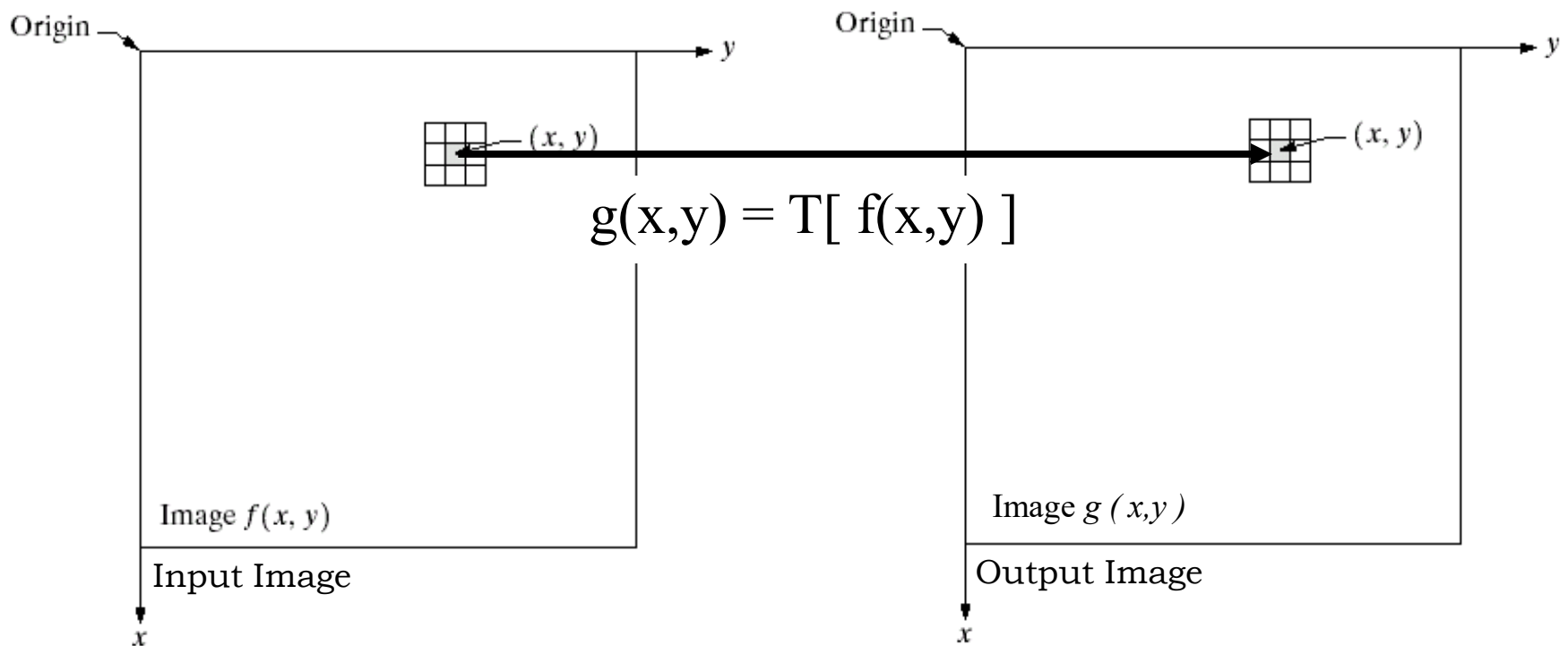
- $f(x,y)$  = input image
- $g(x,y)$  = output, processed, transformed image
- $T$  is an operator on  $f$ 
  - $T$  = intensity transformation function
  - $T$  is defined over some neighborhood of  $(x, y)$ , e.g., a square (3x3) or rectangular sub-image area or a point of 1x1 size
  - $T$  can be operated on a single image or a set of images.

# Spatial Domain



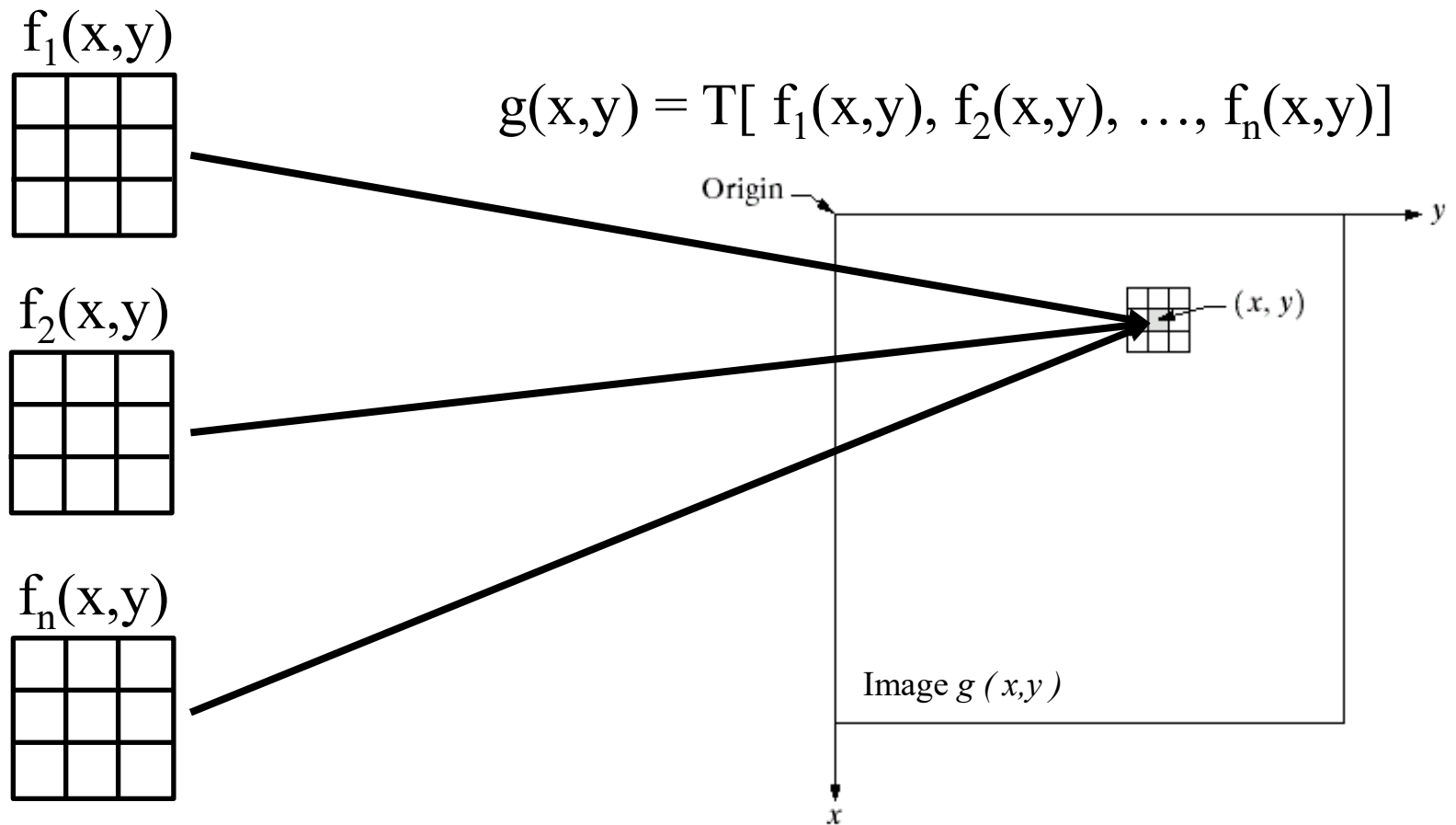
**FIGURE 3.1**

A  $3 \times 3$  neighborhood about a point  $(x_0, y_0)$  in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image. Recall from Chapter 2 that the value of a pixel at location  $(x_0, y_0)$  is  $f(x_0, y_0)$ , the value of the image at that location.



**FIGURE 3.1** A  
 $3 \times 3$   
 neighborhood  
 about a point  
 $(x, y)$  in an image.

$T$  = intensity transformation function



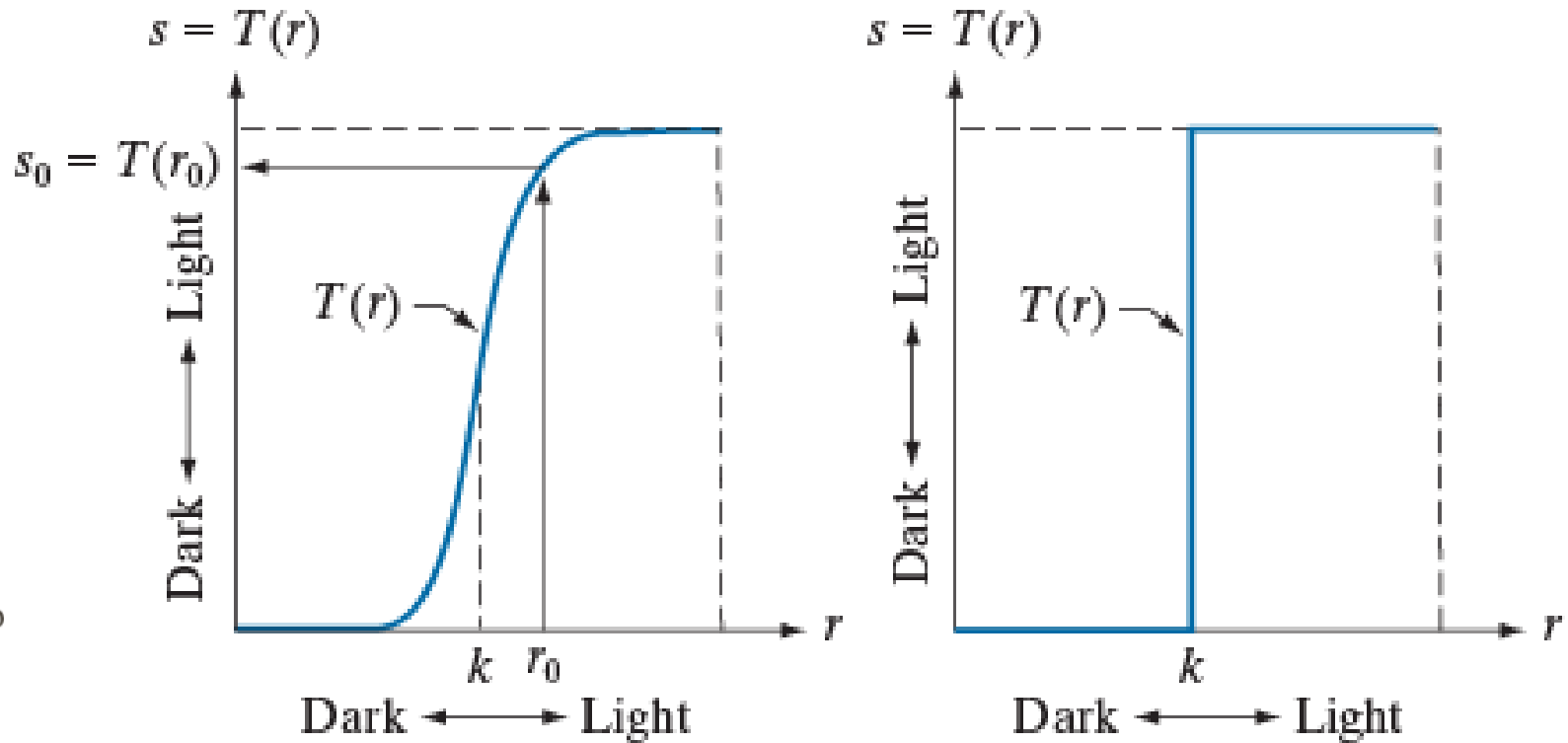
**FIGURE 3.1** A  $3 \times 3$  neighborhood about a point  $(x, y)$  in an image.

$T$  = intensity transformation function

# Gray-level transformation

- Simplest form of  $T$ 
  - neighborhood is  $1 \times 1$  (one pixel)
- Notation:  $s = T(r)$ 
  - $r, s$  denote the gray levels of  $f(x,y)$  and  $g(x,y)$  for any point  $(x,y)$ , respectively

# Examples of $T(r)$



a b

**FIGURE 3.2**

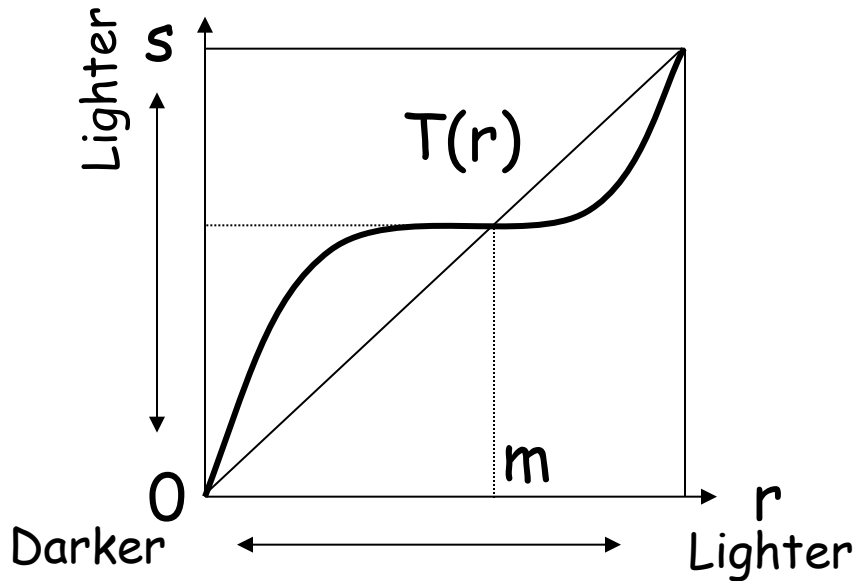
Intensity transformation functions.

(a) Contrast-stretching function.

(b) Thresholding function.

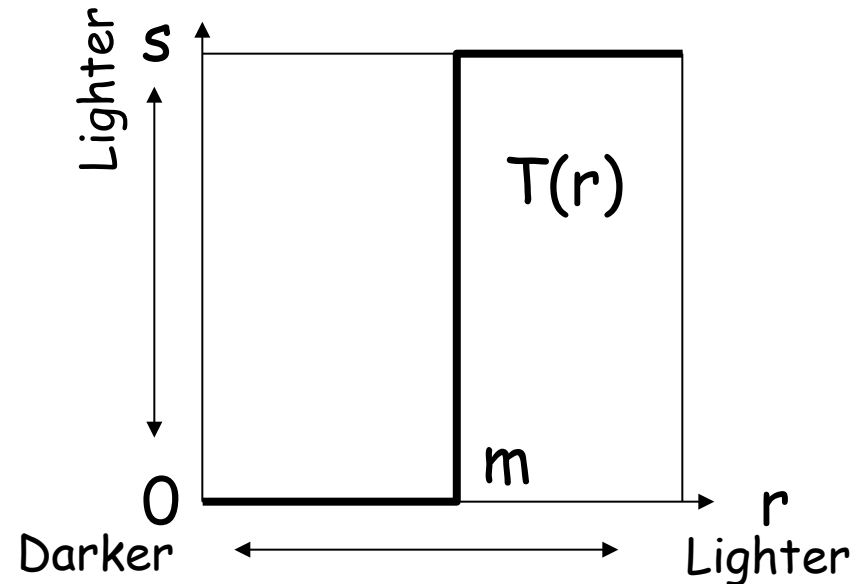


# Examples of $T(r)$



It produces an image of lower contrast than the original by

- (1) darkening the intensity levels above  $m$ , and
- (2) brightening the levels below  $m$  in the original image.

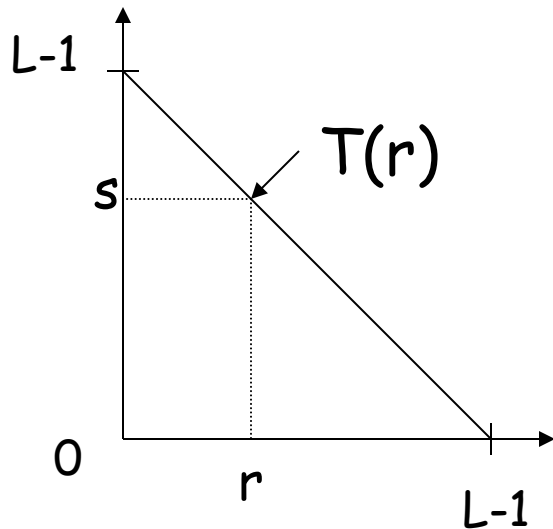


$T$  is a thresholding function

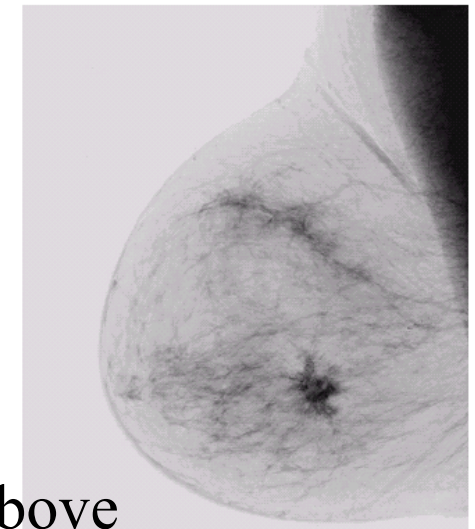
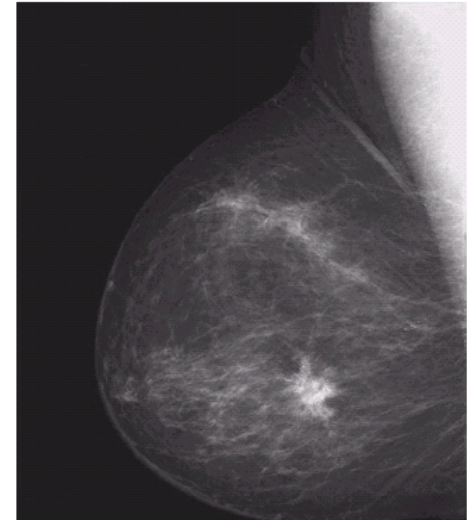
It produces a binary image with an intensity threshold at  $m$ .

Low intensity levels below  $m$  map to zero and high intensity levels above  $m$  map to  $s$ .

# Basic Gray Level Transforms: Inverse or Image Negatives

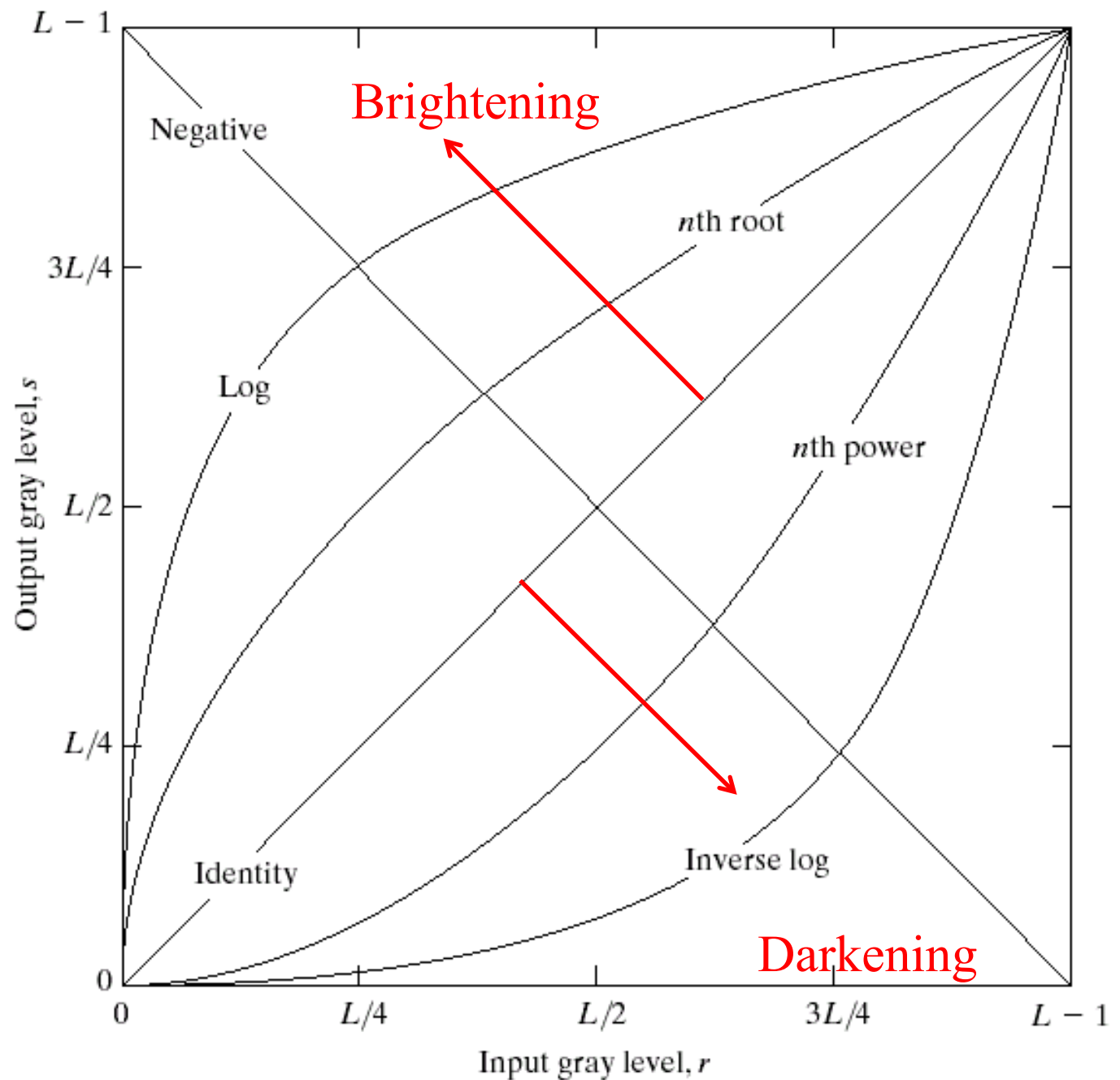


$$s = T(r) = (L-1) - r$$

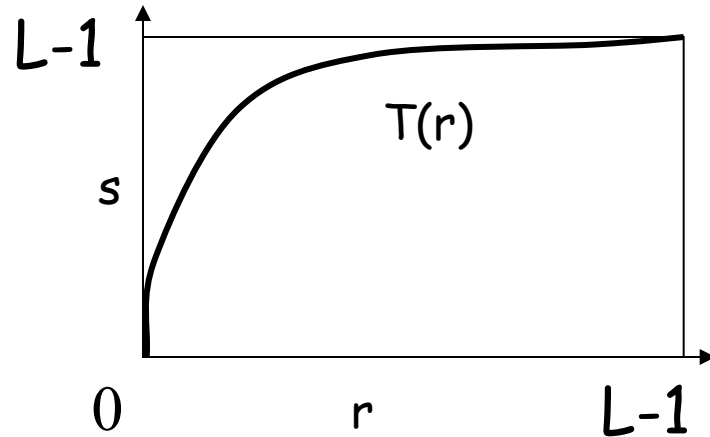


Negatives of the images above

**FIGURE 3.3** Some basic gray-level transformation functions used for image enhancement.



# Log Transformations



$$s = c \log(1 + r)$$

( $c$  is some scale-factor)  
(intensity rescaling maybe needed for  $s$  to fit  $s$  into the range  $[0, L-1]$ )

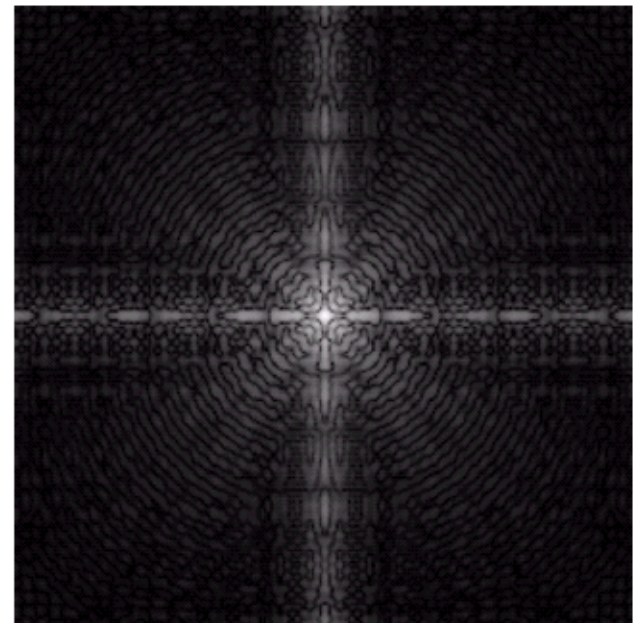
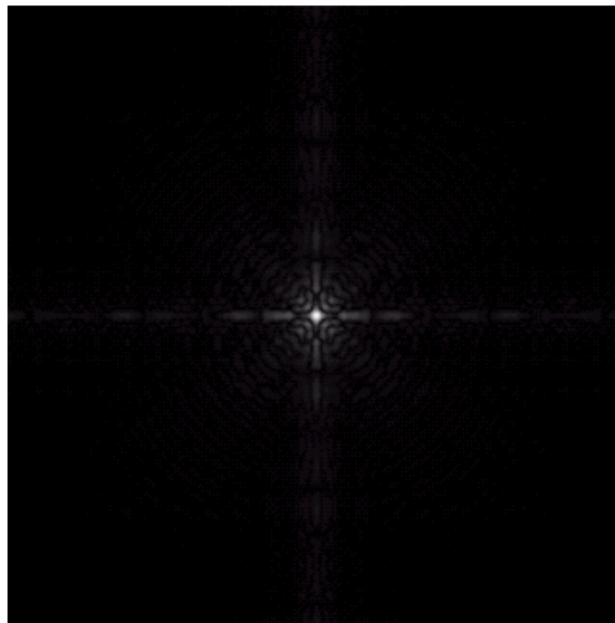
The image becomes brighter and details are more visible.

a b

**FIGURE 3.5**

(a) Fourier spectrum.

(b) Result of applying the log transformation given in Eq. (3.2-2) with  $c = 1$ .



# Power-Law (Gamma) Transformations

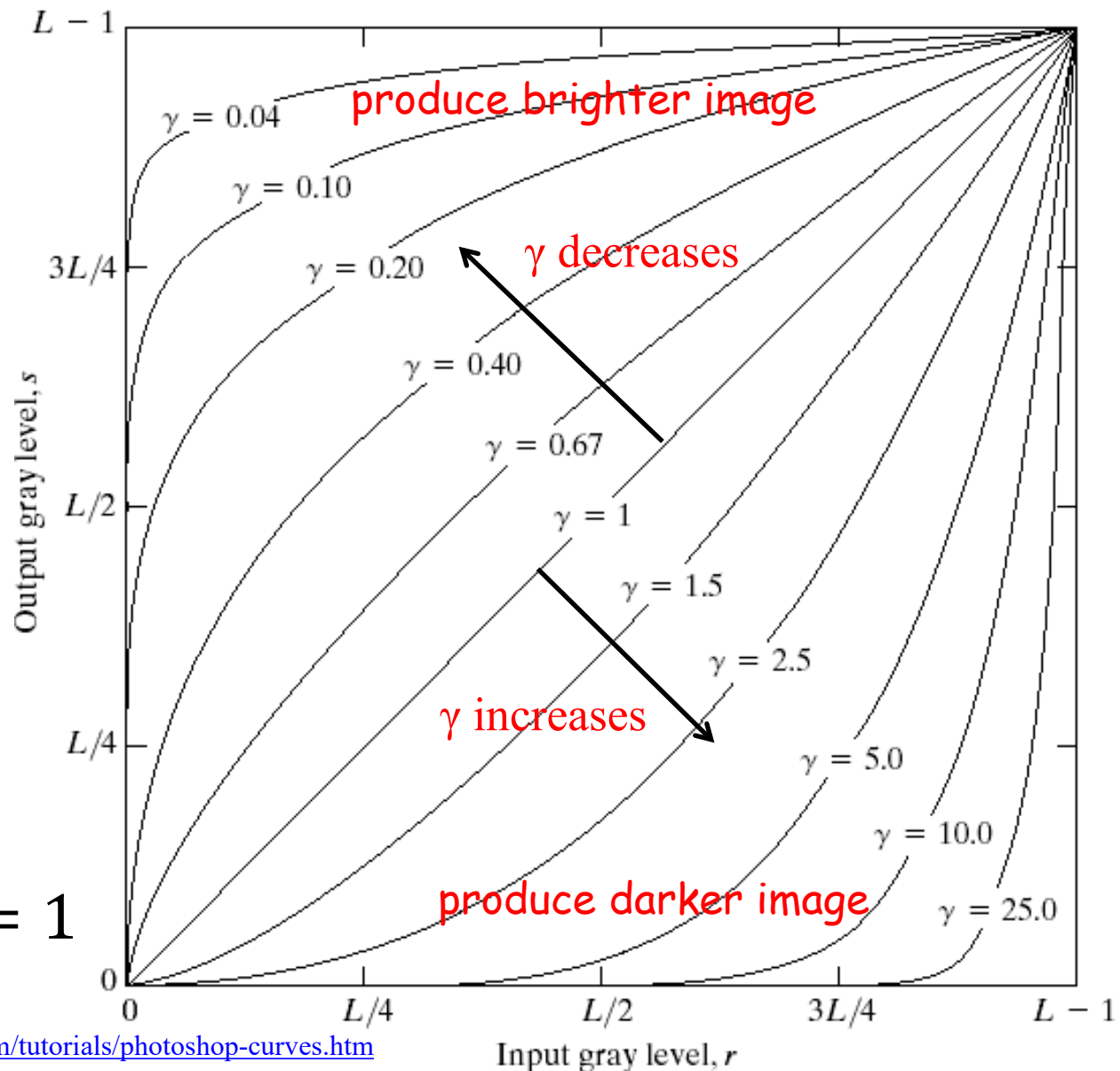
$$s = cr^\gamma$$

where  $C$  and  $\gamma$  are positive constants.

Intensity rescaling may be needed for  $s$  to fit  $s$  into the range  $[0, L-1]$ .

Also, known as *gamma* ( $\gamma$ ) *correction*.

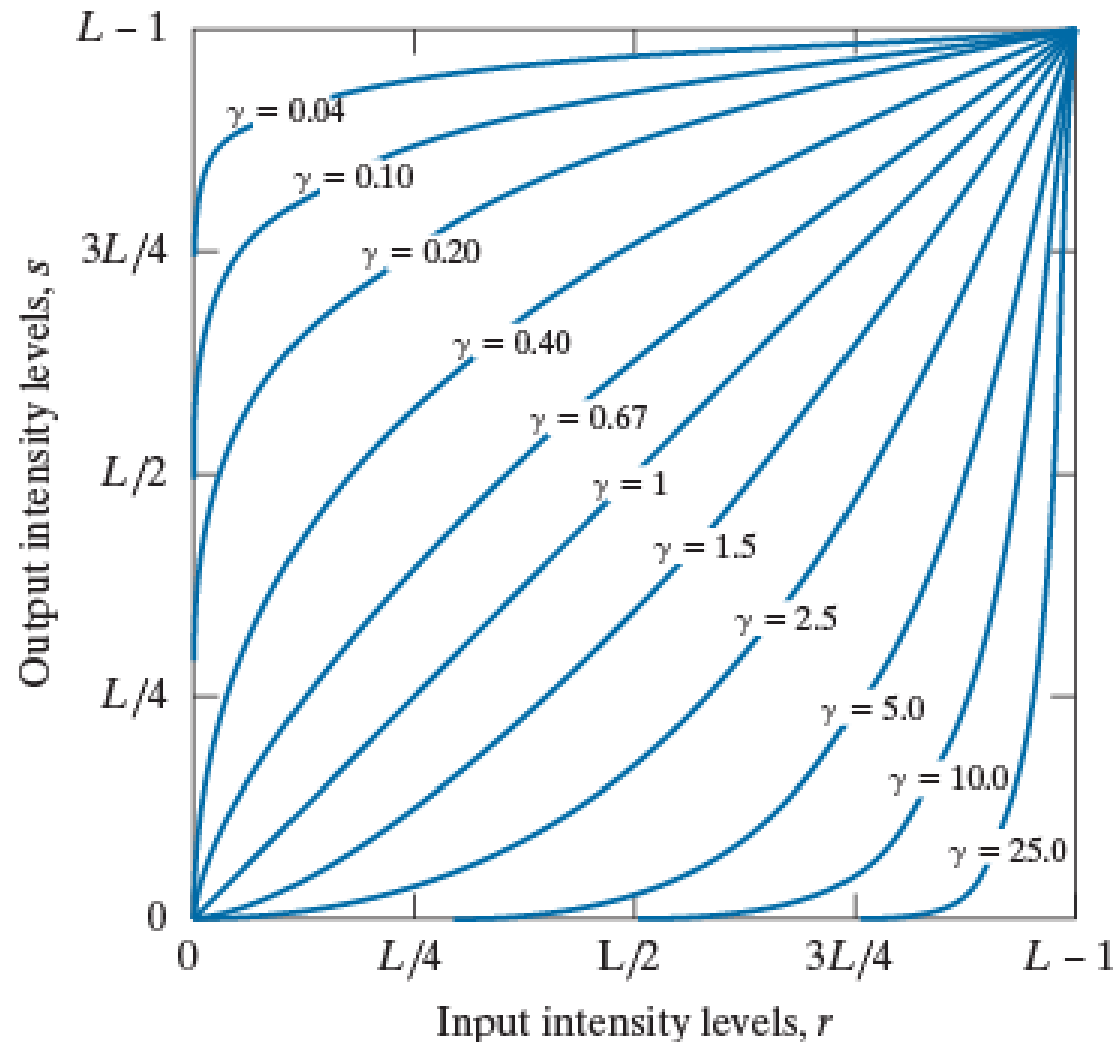
$$C = 1$$



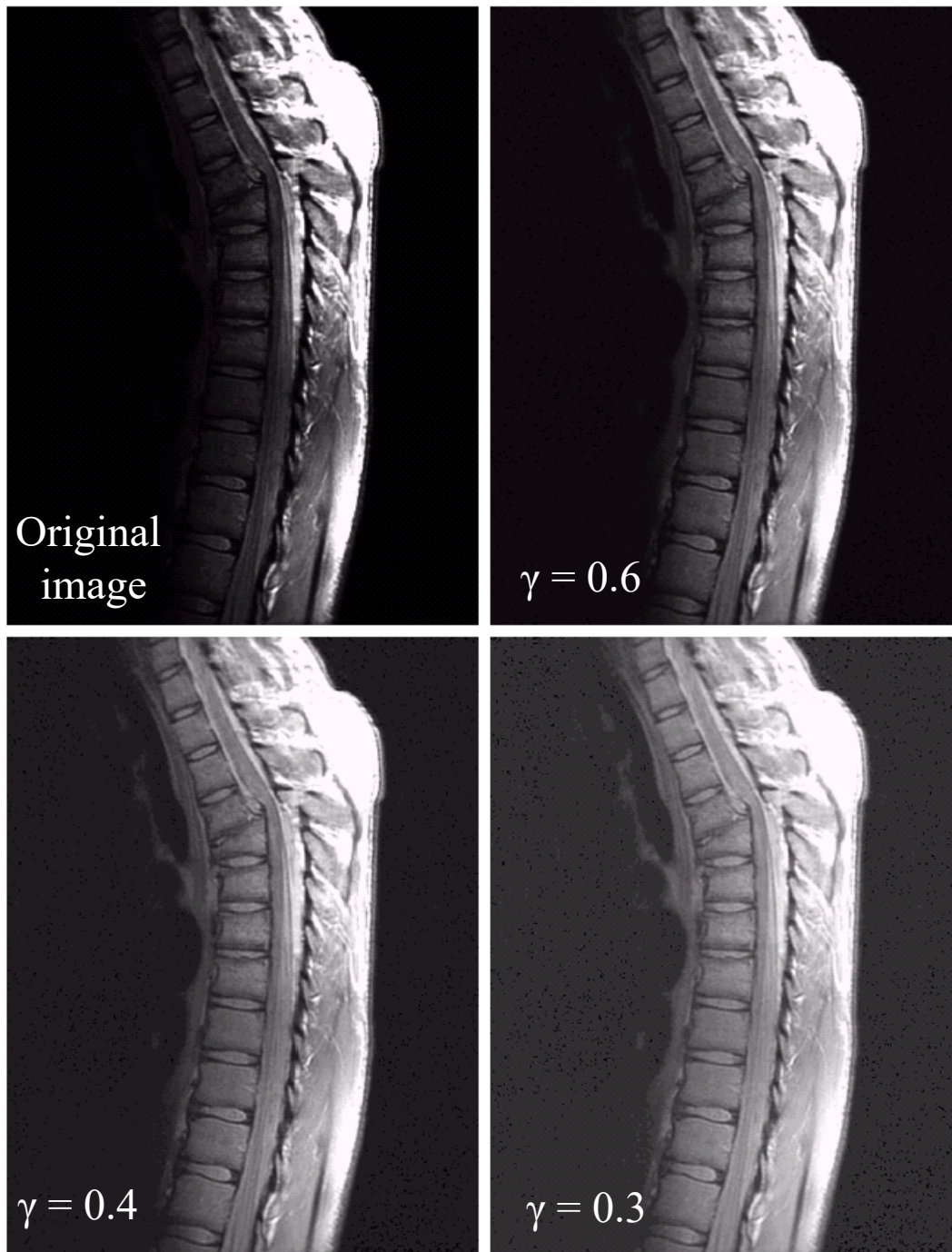
# Power-Law (Gamma) Transformations

**FIGURE 3.6**

Plots of the gamma equation  $s = cr^\gamma$  for various values of  $\gamma$  ( $c = 1$  in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.







|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 3.8**

(a) Magnetic resonance (MR) image of a fractured human spine. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with  $c = 1$  and  $\gamma = 0.6, 0.4$ , and  $0.3$ , respectively. (Original image for this example courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

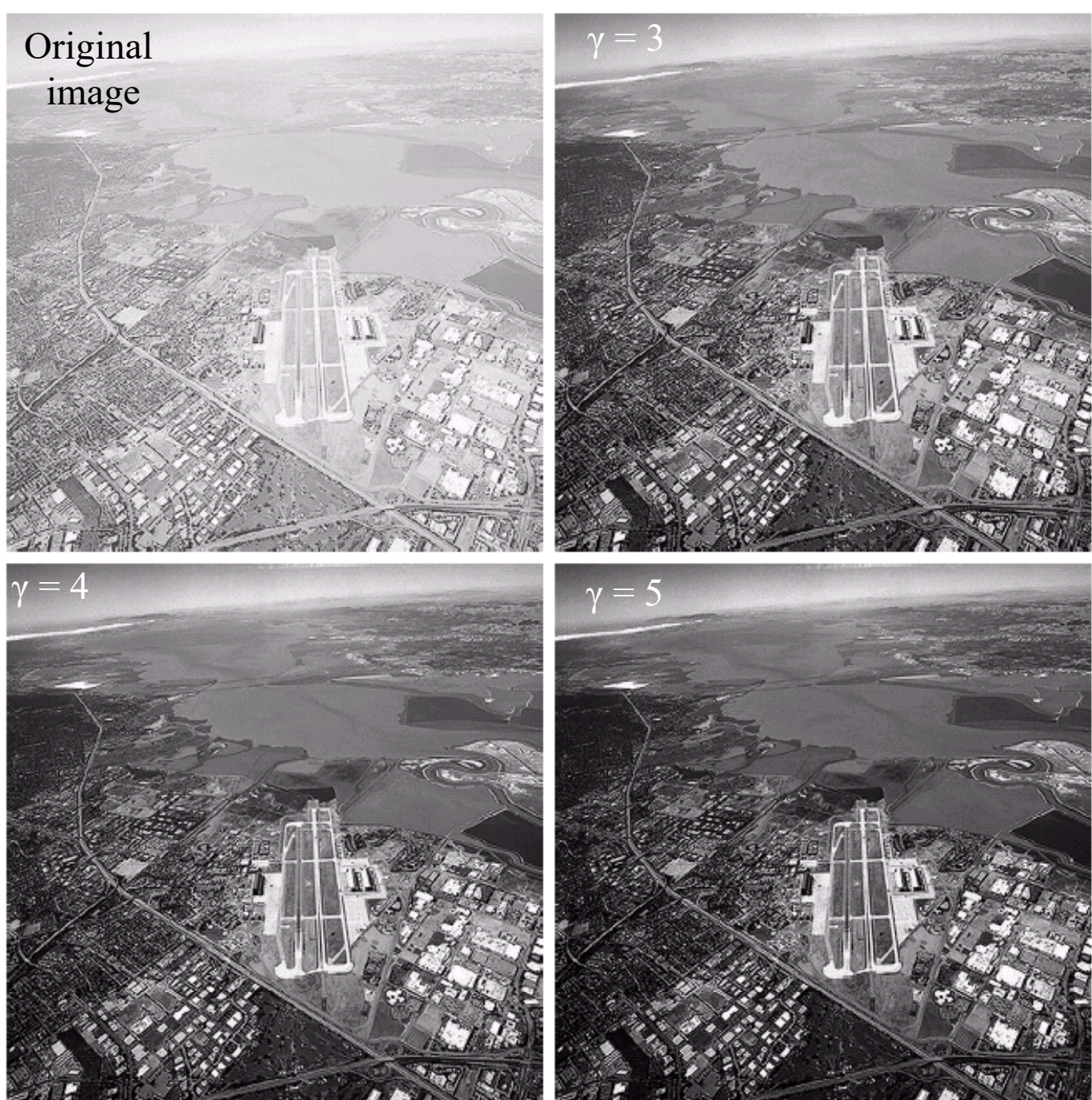
With the help of power-law transformation, fractures near the vertical centre of the spine are now made more visible in the magnetic resonance image.



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 3.9**

(a) Aerial image.  
 (b)–(d) Results of applying the transformation in Eq. (3.2-3) with  $c = 1$  and  $\gamma = 3.0, 4.0$ , and  $5.0$ , respectively. (Original image for this example courtesy of NASA.)

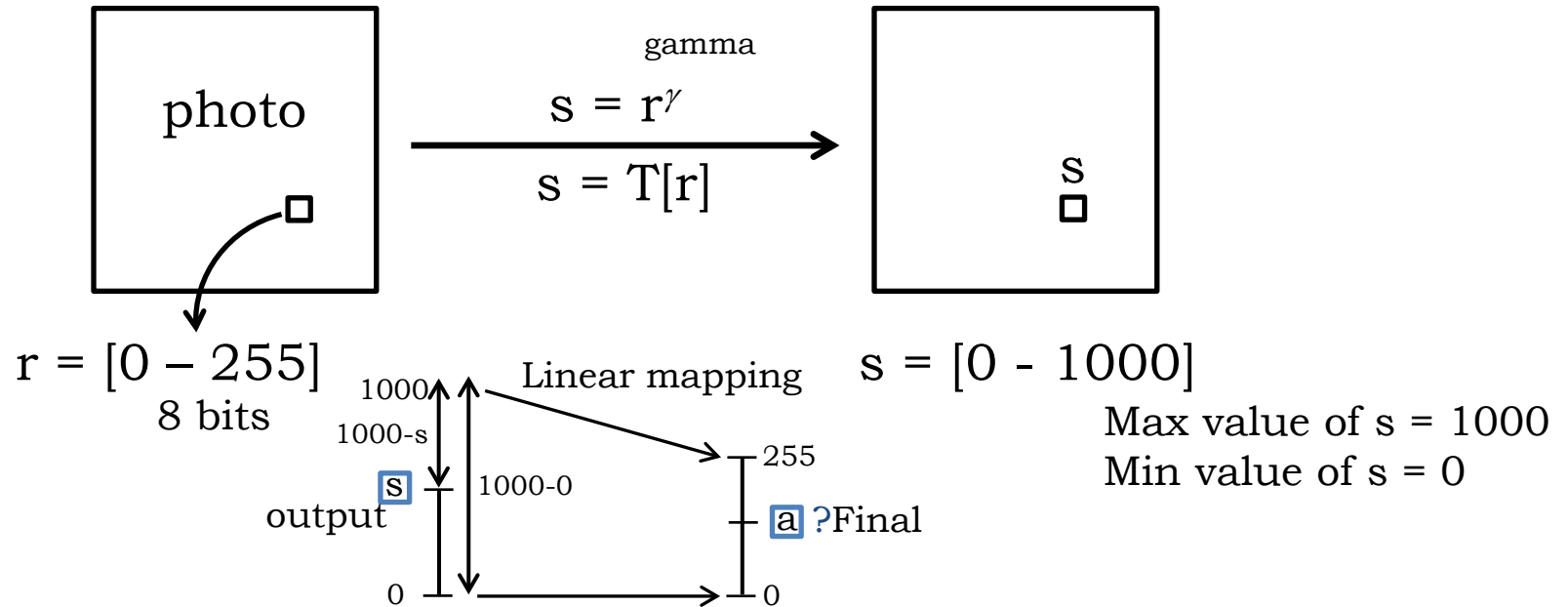




# Intensity normalization

$r$  = input intensity

$s$  = output



Ratio

$$\frac{1000 - s}{1000 - 0} = \frac{255 - a}{255 - 0}$$

(Full range) (Full range)

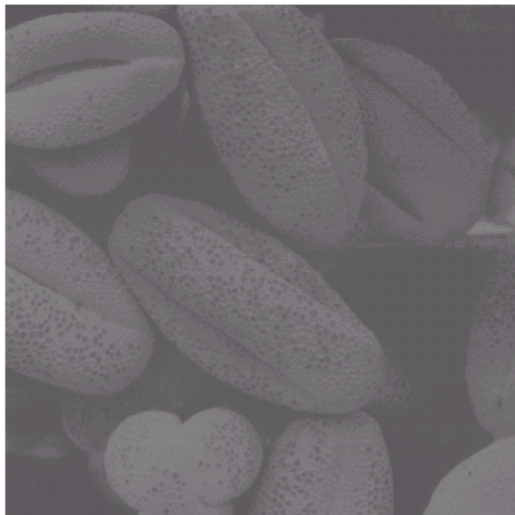
known unknown (Final)

8 bits

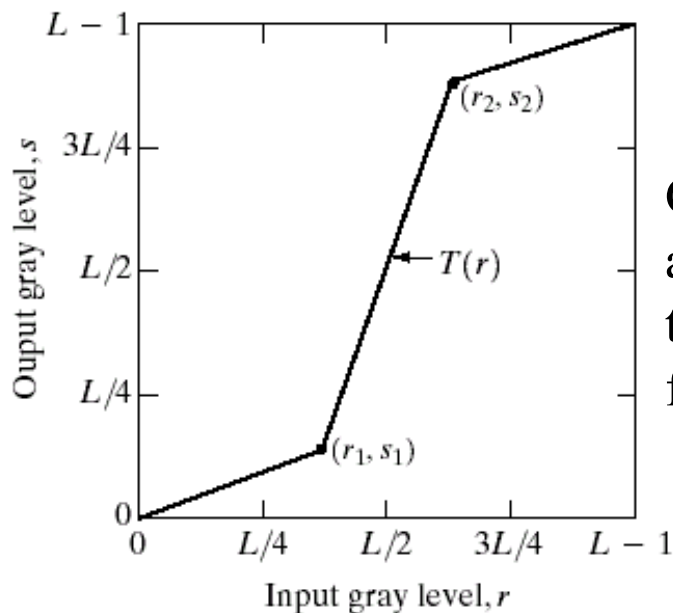
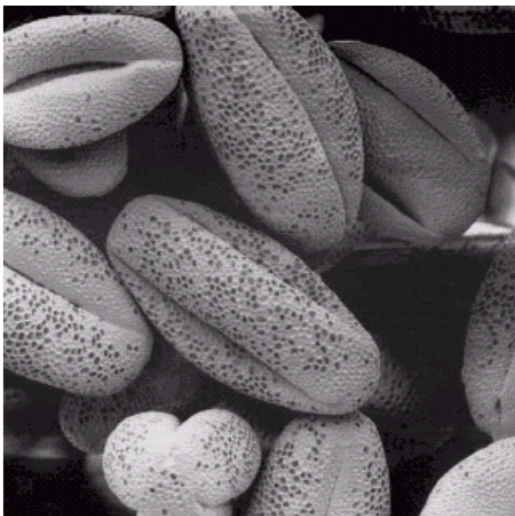
$$a = 255 - \frac{255(1000 - s)}{1000} \quad a = [0 - 255]$$

# Contrast Stretching

A low-contrast image



Output image



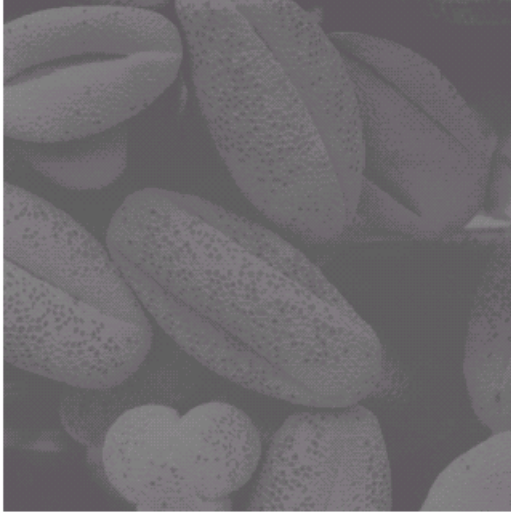
Contrast stretching by  
a piecewise linear  
transformation  
function

$$\begin{aligned} \text{If } r_1 &= r_{\min} & s_1 &= 0 \\ r_2 &= r_{\max} & s_2 &= L-1 \end{aligned}$$

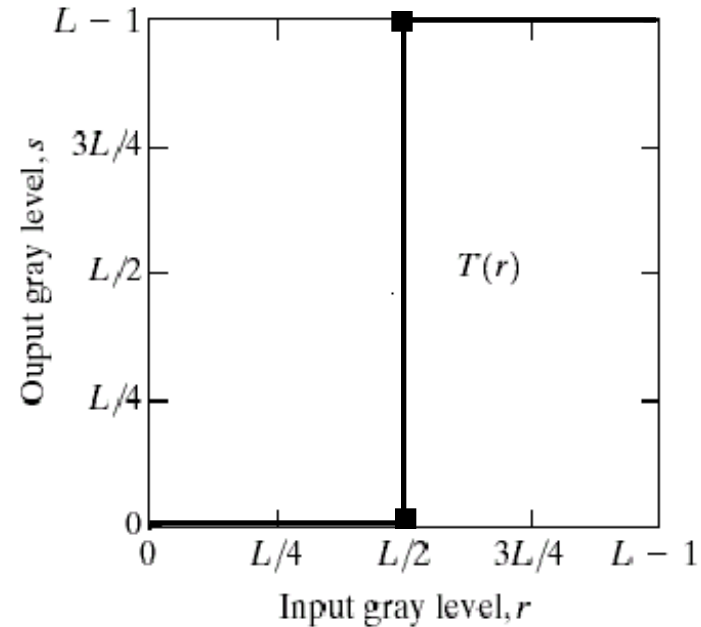
then intensity levels are stretched to the  
full intensity range  $[0, L-1]$ .

# Contrast Stretching

A low-contrast image



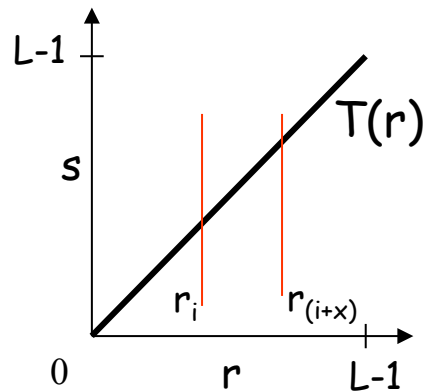
Output binary image



$$\text{If } r_1 = r_2 \quad s_1 = 0 \quad s_2 = L-1$$

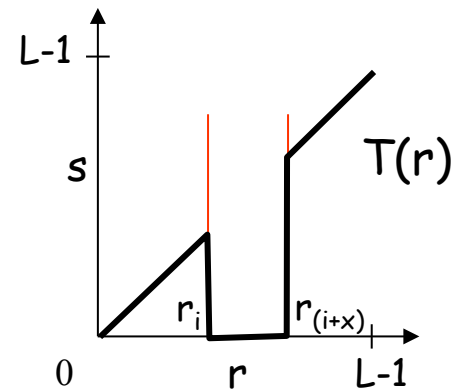
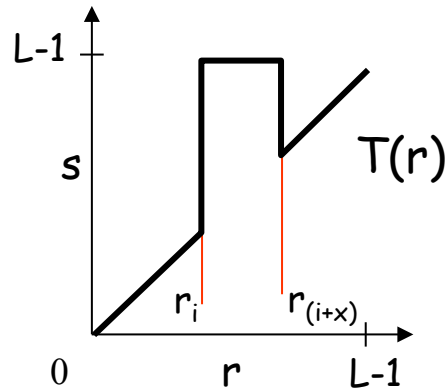
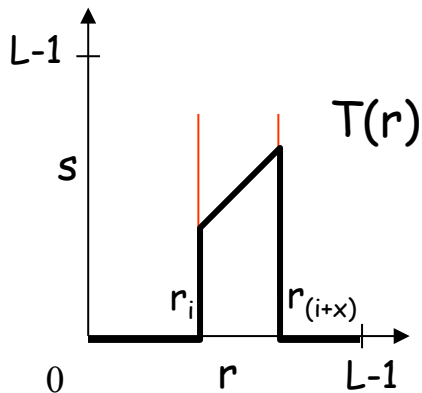
then the image is thresholded and a binary image is produced.

# Windowing (Gray Level Slicing / Intensity Level Slicing)

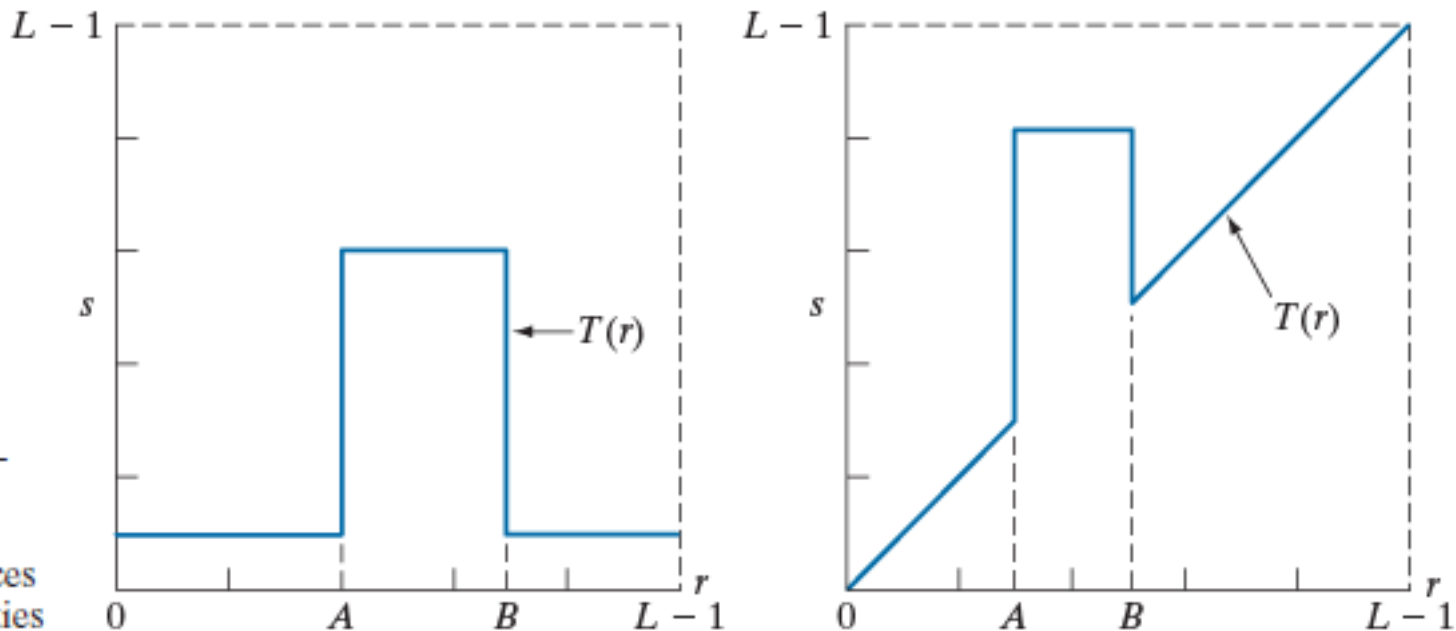


Operating with a "window" of gray levels  $r_i$  to  $r_{(i+x)}$

This function highlights the intensity levels between the range



# Intensity-level Slicing



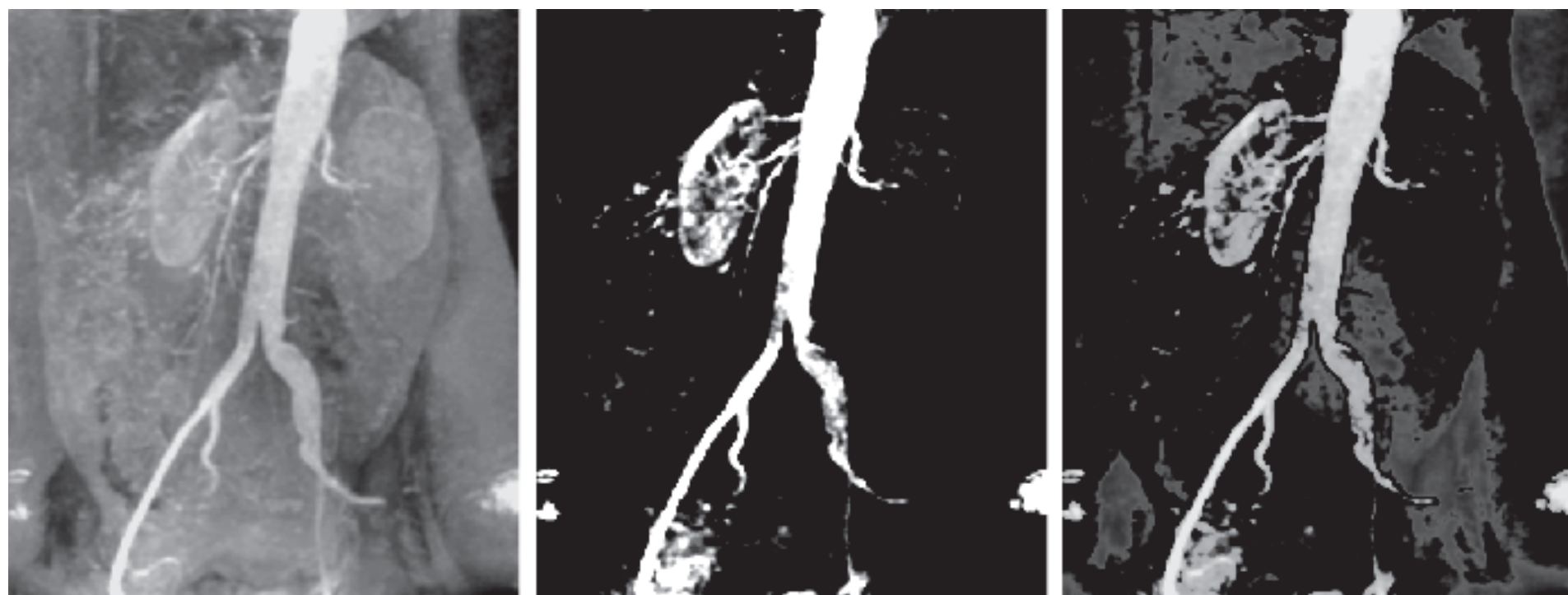
a b

**FIGURE 3.11**

(a) This transformation function highlights range  $[A, B]$  and reduces all other intensities to a lower level.

(b) This function highlights range  $[A, B]$  and leaves other intensities unchanged.

# Intensity-level Slicing

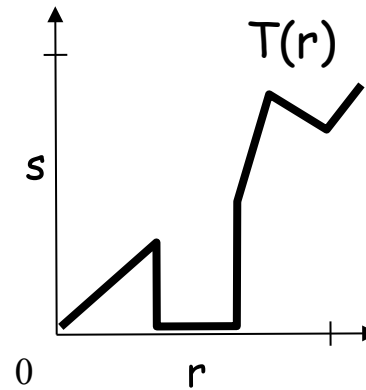


a b c

**FIGURE 3.12** (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

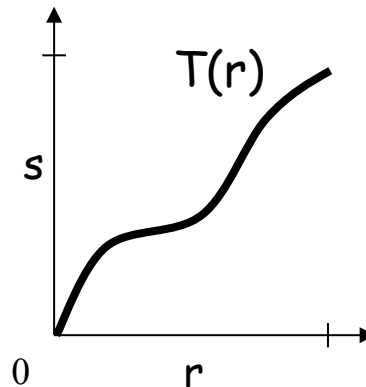
# Properties of $T(r)$

- From  $r$  to  $s$ ,  
one-to-one mapping
- From  $s$  to  $r$ ,  
one-to-many mapping



- No-inverse due to ambiguity
- Does not preserve gray level ordering
- For non-inverse transformation, it is not a one-to-one mapping and is one-to-many mapping.

- One-to-one mapping
- Increasing

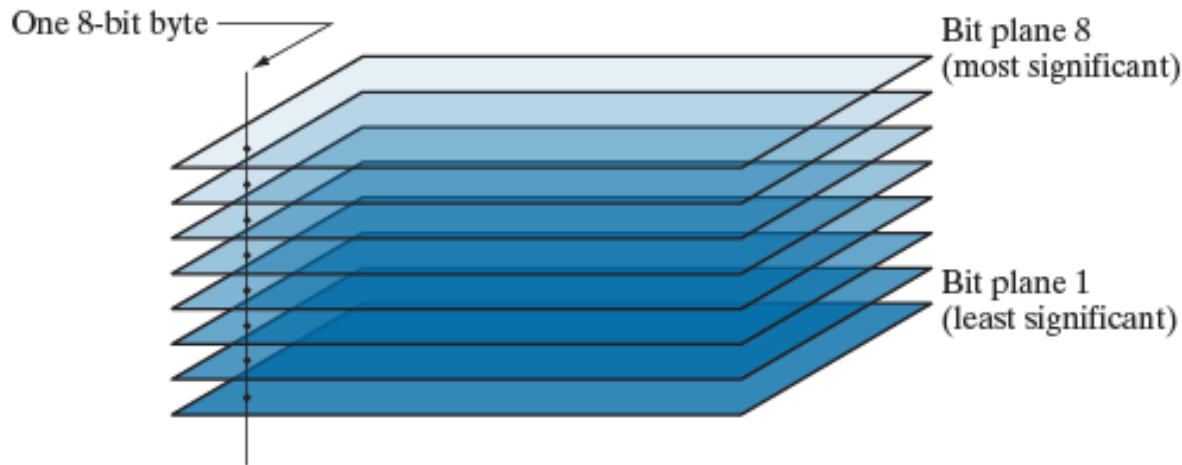


- Inverse
- Preserves gray level ordering (increasing)



# Bit-Plane Slicing

- Pixel values are integers composed of bits
- 8 bits give the values between 0 and 255
- Each bit contributes to the entire image appearance
- Each bit can be viewed as a one-bit image



**FIGURE 3.13**  
Bit-planes of an  
8-bit image.

# Bit-Plane Slicing

- Plane 8 1-bit image is the highest-order bit, most-significant bit, which contains the most visually significant information.
- Decomposing an image into bit planes helps analyze the image properties

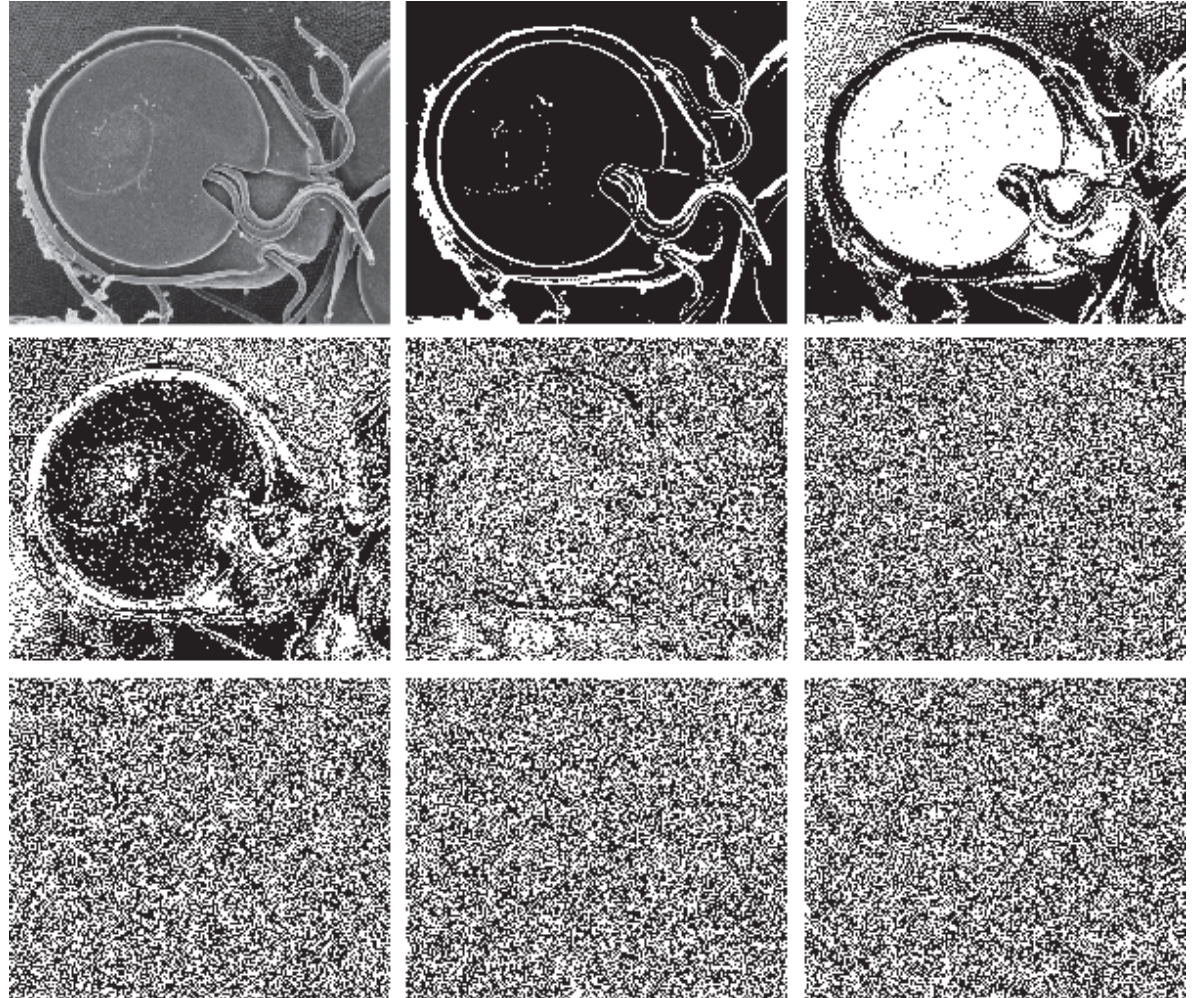
|   |   |   |
|---|---|---|
| a | b | c |
| d | e | f |
| g | h | i |

**FIGURE 3.14**

(a) An 8-bit gray-scale image of size  $837 \times 988$  pixels.

(b) through (i) Bit planes 8 through 1, respectively, where plane 1 contains the least significant bit. Each bit plane is a binary image.

Figure (a) is an SEM image of a trophozoite that causes a disease called *giardiasis*. (Courtesy of Dr. Stan Erlandsen, U.S. Center for Disease Control and Prevention.)

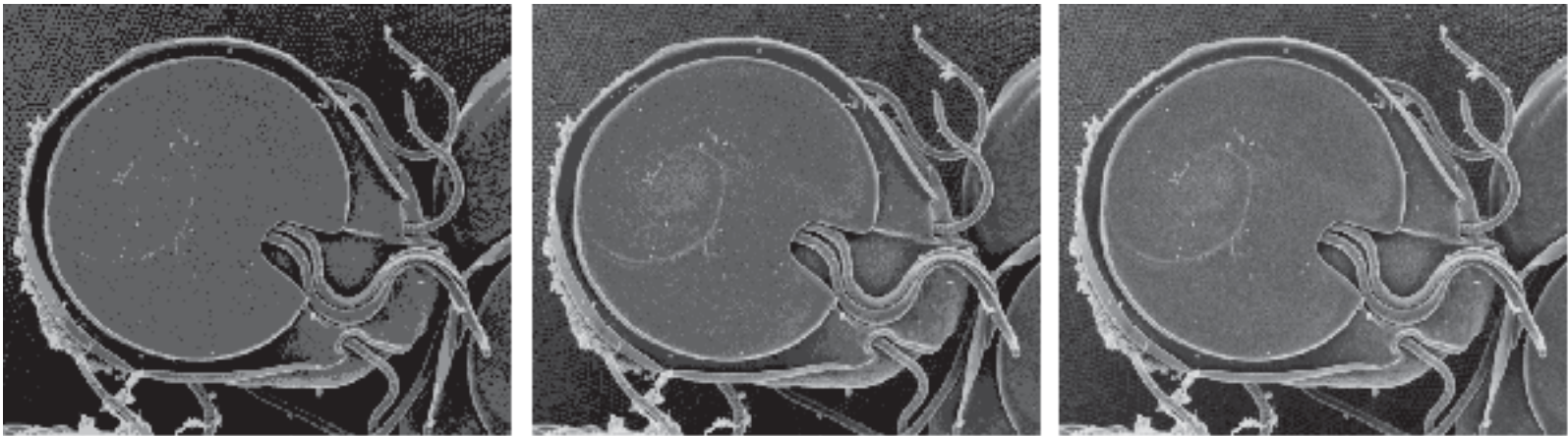


# Bit-Plane Slicing

- An image can be reconstructed by combining different bit planes
- The  $n^{\text{th}}$  bit plane times  $2^{n-1}$  is added to the reconstructed image to form the grey scale image
- If there are  $n$  bits, the constructed image has  $2^n$  image intensity levels.

a b c

**FIGURE 3.15**  
Image  
reconstructed  
from bit planes:  
(a) 8 and 7;  
(b) 8, 7, and 6;  
(c) 8, 7, 6, and 5.





# Digital Watermarking

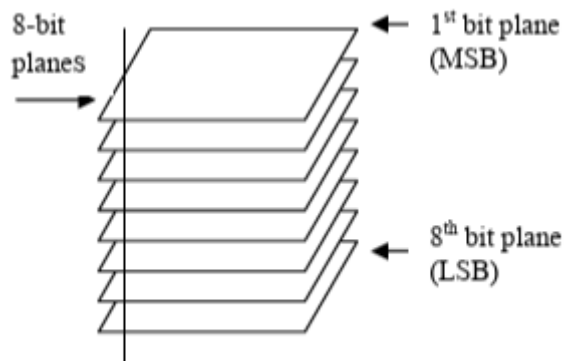
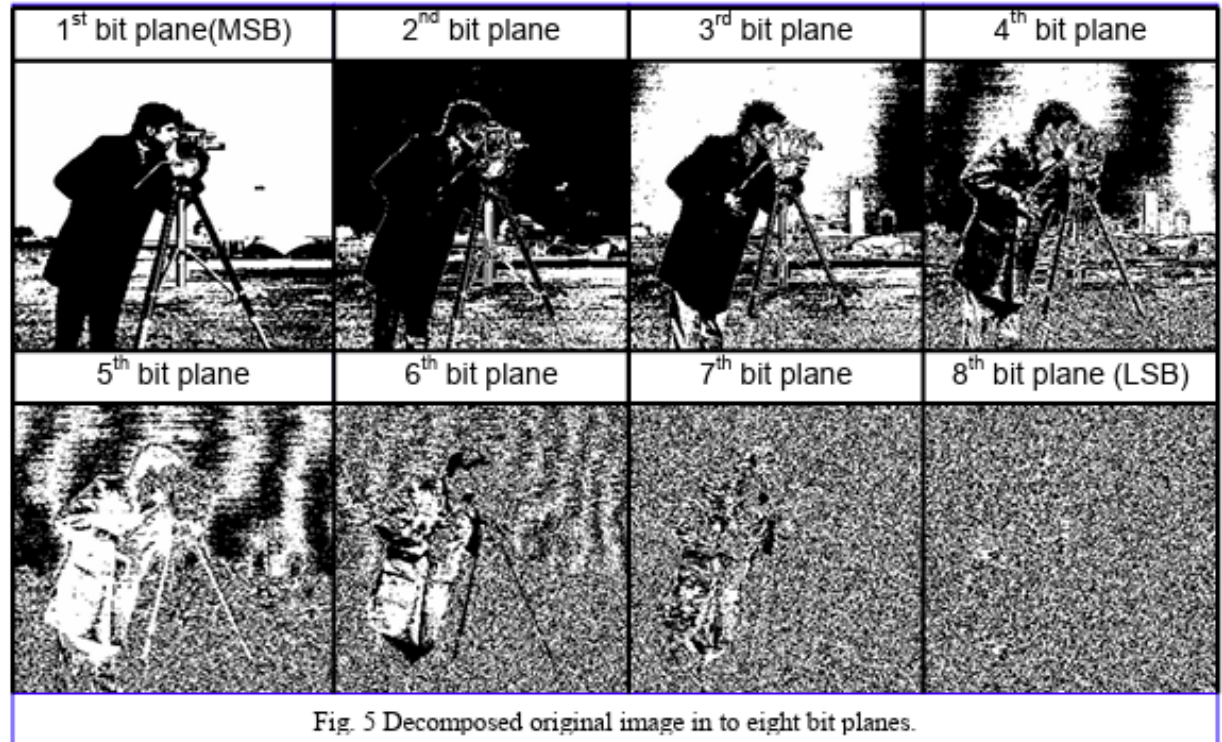


Fig. 3. Bit plane representation of an image



# Digital Watermarking

